eleVADR Detection Module Reference

Compact technical guide to the 75 detection modules, their inputs, limitations, and non-default Zeek requirements.

Important: Zeek is the network telemetry engine; it generally does not issue the eleVADR finding itself. Zeek protocol analyzers turn packets into structured logs. eleVADR normalizes those logs into an AnalysisContext, combines them with the Detection Context (assets, segments, trusted infrastructure, authorization and allowlists), then each detector applies its rule. Observed Zeek traffic is evidence only — it does not create authorization or policy.

Bottom line: all 75 modules below are accurate — no module claims a log, field, or capability that does not actually exist, and every description matches what the code reads and what Zeek can produce. 73 of the 75 are possible outright (some only with the extra packages, scripts, or Zeek versions called out below). Two modules (**kerberos_asrep_roastable_accounts**, **smb_signing_disabled_ntlmv1**) are **not possible from any stock Zeek log** and are therefore **disabled by default** — flagged in red below — so a run never silently claims coverage it cannot deliver.

What changed in this edition: every module entry below states its actual log dependency as verified against the detector source code (not the originally documented dependency list, which in several cases referenced logs that do not exist in any Zeek distribution, such as smb.log and ldap_bind.log). Each entry also carries a **Limitations** note (tan) describing what the module cannot prove from its stated evidence, and a **Non-default Zeek requirement** note naming any log that needs an ICSNPP package, another add-on package, a custom script, or a non-default Zeek version/configuration before the module can produce findings.

Zeek log-source tiers used in this reference

Log(s)	Tier	What it needs
conn.log, dns.log, http.log, ssl.log, x509.log, ftp.log, smtp.log, ssh.log, ntlm.log, kerberos.log, rdp.log, dhcp.log, ntp.log, snmp.log, weird.log, files.log, socks.log, smb_mapping.log, smb_files.log, dce_rpc.log	Stock Zeek (default)	Ships enabled by default; no package or script needed.
ldap.log, ldap_search.log, quic.log	Stock Zeek (version-gated)	Requires Zeek 6.1 or newer.
modbus.log, dnp3.log	Stock Zeek (limited fields)	Analyzer ships and is on by default, but only exposes baseline function-code fields; ICSNPP extension scripts add richer profiling.
smb_cmd.log	Stock Zeek, disabled by default	Must be explicitly enabled (SMB::logged_smb_cmds); debug-oriented, drops commands on busy streams.
enip.log, cip.log, bacnet.log, bacnet_discovery.log, s7comm.log, mms.log	ICSNPP package required	CISA/INL packages installed via zkg; absent from stock Zeek.
JA3/JA3S fields in ssl.log	Add-on package required	Base ssl.log has no JA3 field; a separate JA3 package must be loaded.
vnc.log, tftp.log, upnp.log, upnp_igd.log, ssdp.log	Add-on package required	Separate Zeek packages; not shipped with stock Zeek.
arp.log, telnet credential capture, legacy login.log	Custom script required	No stock log exists; needs a hand-written Zeek script, and ARP additionally requires on-segment sensor placement.
smb.log, ldap_bind.log	Not a real Zeek log	These names do not exist in any Zeek distribution; do not budget for them.
IEC 61850 GOOSE / Sampled Values	Not obtainable from Zeek	L2 multicast, non-IP traffic; invisible to an IP-oriented Zeek sensor regardless of packages installed.

Reading each entry: Inputs lists the Zeek logs a module hard-requires, any "one of" alternative set, and any log that only sharpens confidence/timing/granularity when present. Detection states what eleVADR evaluates using that evidence. A tan box appears under an entry only when the module carries a limitation or needs something beyond a stock Zeek install. A red box appears only on the two modules that are disabled by default because stock Zeek cannot supply their evidence at all.

01. ARP / IP-MAC Identity Change

`arp_ip_mac_identity_change`

Inputs: arp.log.

Detection: Detects ARP observations where an authoritative asset IP is claimed by an unexpected MAC address, or where one IP rapidly appears behind multiple MAC addresses.

ARP visibility depends on capture placement.

Non-default Zeek requirement: arp.log — Custom script required: No stock arp.log. Requires a custom ARP-logging script, and the sensor must sit on the Layer-2 broadcast segment to see the traffic at all.

02. ARP / L2 Reconnaissance

`arp_l2_reconnaissance`

Inputs: arp.log.

Detection: Detects hosts rapidly probing many distinct IPv4 targets with ARP requests.

ARP visibility depends on capture placement.

Non-default Zeek requirement: arp.log — Custom script required: No stock arp.log. Requires a custom ARP-logging script, and the sensor must sit on the Layer-2 broadcast segment to see the traffic at all.

03. BACnet Who-Is/I-Am Flood or External Discovery

`bacnet_discovery_anomalies`

Inputs: bacnet.log; also used when present: bacnet_discovery.log.

Detection: Detects abnormal BACnet Who-Is/I-Am discovery bursts and BACnet discovery activity originating from explicitly non-BAS network segments.

Unavailable without protocol-specific BACnet evidence.

Non-default Zeek requirement: bacnet.log — ICSNPP package required: Requires the ICSNPP-BACnet package (zkg install icsnpp-bacnet). Not present in stock Zeek. | bacnet_discovery.log — ICSNPP package required: Requires the ICSNPP-BACnet package. Not present in stock Zeek.

04. Beaconing / C2 Communication

`beaconing_c2_communication`

Inputs: conn.log.

Detection: Identifies repeated outbound connections to the same destination at consistent intervals, a behavioral pattern commonly associated with malware command-and-control check-ins.

Log presence and field coverage must be verified at runtime.

05. Brute Force Authentication

`brute_force_authentication`

Inputs: any of (no single log mandatory): http.log, ssh.log, kerberos.log, ntlm.log, rdp.log, vnc.log, conn.log.

Detection: Identifies repeated authentication failures against SSH, RDP, HTTP, Kerberos, SMB/NTLM, and VNC that are consistent with automated password guessing or credential spraying.

Log presence and field coverage must be verified at runtime.

Non-default Zeek requirement: vnc.log — Add-on package required: Requires a separate Zeek package (not in stock Zeek).

06. Cleartext Credentials

`cleartext_credentials`

Inputs: any of (no single log mandatory): ftp.log, http.log, smtp.log.

Detection: Identifies cleartext credential exposure in FTP, HTTP, and SMTP analysis data. Password values are redacted from results.

Telnet support was removed from this module. Modern Zeek ships no credential analyzer for Telnet payload content, so there was no log for that half of the module to read from.

07. CoDeSys Runtime Exposure

`codesys_runtime_exposure`

Inputs: conn.log.

Detection: Identifies CODESYS programming/runtime and gateway communications using vendor-documented default ports or explicit service labels, with higher severity when runtime access reaches configured OT controllers from public or non-OT sources.

Log presence and field coverage must be verified at runtime.

08. Control System to Enterprise IT Communication (Non-DMZ)

`control_system_enterprise_non_dmz`

Inputs: conn.log.

Detection: Identifies direct communication between configured control-system/OT assets and enterprise or other non-OT networks when neither observed endpoint is an approved control-system DMZ intermediary.

Log presence and field coverage must be verified at runtime.

09. Controller Communication Jitter

`controller_communication_jitter`

Inputs: conn.log; optional (sharpens confidence/timing when present): modbus.log, dnp3.log, s7comm.log, enip.log, cip.log.

Detection: Detects increased timing variance in established controller communications after a baseline. Uses per-message OT protocol-log timing when it is more granular than conn.log, so jitter inside one persistent session is still visible.

Long-lived TCP sessions hide individual polling intervals.

Non-default Zeek requirement: s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek. | enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek. | cip.log — ICSNPP package required: Requires the ICSNPP-ENIP package. ICSNPP-ENIP writes CIP service-code detail to a separate cip.log, not enip.log. Not present in stock Zeek.

10. Cross-Purdue-Level Traffic

`cross_purdue_level_traffic`

Inputs: conn.log.

Detection: Identifies direct communications between Purdue levels that should not communicate directly, including control levels reaching enterprise levels without an intermediate industrial DMZ.

Log presence and field coverage must be verified at runtime.

11. Database Service Exposed

`database_service_exposed`

Inputs: conn.log.

Detection: Identifies database services reachable across configured network-segment boundaries or from public Internet addresses, including MySQL, SQL Server, PostgreSQL, MongoDB, Redis, Memcached, and Elasticsearch.

Log presence and field coverage must be verified at runtime.

12. Deprecated / Insecure Connection Services and Protocols

`deprecated_insecure_services_protocols`

Inputs: conn.log; also used when present: ftp.log, http.log, smtp.log, telnet.log, login.log.

Detection: Identifies observed use of legacy, plaintext, or otherwise insecure connection services and insecure connection protocols.

Log presence and field coverage must be verified at runtime.

Non-default Zeek requirement: telnet.log — Custom script required: Modern Zeek ships no credential-capturing Telnet analyzer by default; there is no reliable stock Telnet credential log. | login.log — Custom script required: Legacy Zeek login framework; not a default/stock output on modern Zeek.

13. PPTP / Deprecated VPN Protocol

`deprecated_vpn_protocol`

Inputs: conn.log.

Detection: Identifies PPTP using TCP/1723 with GRE when available, and L2TP sessions on UDP/1701 that lack correlated IKE/IPsec evidence.

Log presence and field coverage must be verified at runtime.

14. DNS Resolver Source Drift

`dns_source_drift`

Inputs: dns.log.

Detection: Detects OT clients that use DNS resolvers outside explicitly trusted infrastructure or change resolver after a capture baseline.

Log presence and field coverage must be verified at runtime.

15. DNS Tunneling / Exfiltration

`dns_tunneling_exfiltration`

Inputs: dns.log.

Detection: Identifies DNS behavior consistent with covert tunneling or exfiltration, including encoded/high-entropy subdomains, high-payload TXT/NULL/ANY queries, and sustained NXDOMAIN-heavy query patterns.

Log presence and field coverage must be verified at runtime.

16. Encrypted Session Fingerprint Change

`encrypted_session_fingerprint_change`

Inputs: ssl.log.

Detection: Detects established TLS endpoint pairs whose observed TLS fingerprint changes after a capture baseline.

JA3-specific drift requires extension fields.

17. Engineers Tools Over Clear-Text (TFTP/FTP)

`engineering_tools_cleartext`

Inputs: one of: ftp.log, tftp.log, files.log.

Detection: Detects firmware or configuration material transferred using clear-text FTP or TFTP, using Zeek FTP/TFTP/file metadata rather than port-only inference.

non-base/extension logs. Firmware/config identification is inferred from file MIME type/name heuristics and is best-effort, not a reliable determination.

Non-default Zeek requirement: tftp.log — Add-on package required: Requires a separate Zeek Spicy-based package (not in stock Zeek).

18. Engineering Workstation Control Burst

`engineering_workstation_control_burst`

Inputs: one of: modbus.log, dnp3.log, s7comm.log, enip.log.

Detection: Detects bursts of control-write/programming operations from authoritative engineering workstations, including bursts on otherwise authorized paths.

non-base/extension logs

Non-default Zeek requirement: s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek. | enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek.

19. EtherNet/IP CIP Write and Session Abuses

`enip_cip_write_session_abuses`

Inputs: one of: enip.log, cip.log.

Detection: Detects explicit CIP write services and excessive Forward_Open/Large_Forward_Open activity in EtherNet/IP telemetry, with optional policy checks for approved write paths. Reads CIP service-code detail from cip.log when ICSNPP-ENIP is loaded, falling back to enip.log alone when it is not.

ICSNPP-ENIP writes enip.log (session/registration) AND cip.log (CIP service codes); CIP write/Forward_Open detail lives in cip.log, not enip.log. ENIP-only records without cip.log may not establish CIP writes.

Non-default Zeek requirement: enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek. | cip.log — ICSNPP package required: Requires the ICSNPP-ENIP package. ICSNPP-ENIP writes CIP service-code detail to a separate cip.log, not enip.log. Not present in stock Zeek.

20. Excessive Broadcast / Multicast in OT

`excessive_broadcast_multicast_ot`

Inputs: conn.log.

Detection: Detects unusually high broadcast or multicast traffic rates from explicitly configured OT segments/VLANs where point-to-point traffic is expected to dominate. Uses Zeek conn.log packet counts when available and falls back to flow-event rates when packet counts are unavailable.

Log presence and field coverage must be verified at runtime.

21. File Extraction on Wire (Sensitive Types)

`file_extraction_sensitive_types`

Inputs: files.log.

Detection: Detects executable, configuration, or archive file types observed by Zeek files.log when they are transferred to external destinations or across explicitly classified OT/IT boundaries.

Log presence and field coverage must be verified at runtime. Sensitive-type classification is based on MIME type, which is coarse and best-effort; it is not a reliable determination of file sensitivity.

22. High Fan-Out or Fan-In Patterns

`high_fan_in_out`

Inputs: conn.log.

Detection: Identifies bursts where one host contacts many distinct destinations (fan-out), or many distinct sources suddenly contact one newly observed destination (fan-in).

Log presence and field coverage must be verified at runtime.

23. HTTP User-Agent Anomalies

`http_user_agent_anomalies`

Inputs: http.log; also used when present: conn.log.

Detection: Identifies CLI/scripted HTTP clients and rare or missing User-Agent strings on sensitive HTTP flows using protocol telemetry and conservative baselines.

Log presence and field coverage must be verified at runtime.

24. ICCP / Inter-Control-Center Protocol

`iccp_tase2_detected`

Inputs: conn.log; also used when present: mms.log.

Detection: Identifies traffic consistent with ICCP/TASE.2, used for real-time data exchange between electric utility SCADA control centers. Explicit ICCP/TASE.2 service labels are preferred; TCP/102/OSI-on-TCP is treated conservatively because other industrial protocols also use ISO-on-TCP.

mms.log requires the ICSNPP-MMS package; stock Zeek has no MMS/ISO-on-TCP parser. Without it, this module can only see TCP/102 by port, which is treated conservatively because other industrial protocols also use ISO-on-TCP. IEC 61850 GOOSE/Sampled Values are L2 multicast, non-IP traffic and are invisible to an IP-oriented Zeek sensor; only MMS-over-TCP is covered here.

Non-default Zeek requirement: mms.log — ICSNPP package required: Requires the ICSNPP-MMS package (zkg install icsnpp-mms). Not present in stock Zeek. Covers MMS-over-TCP only; IEC 61850 GOOSE/Sampled Values are L2 multicast, non-IP traffic and are not obtainable from any Zeek log.

25. ICMP as Data Channel

`icmp_data_channel`

Inputs: conn.log.

Detection: Identifies sustained ICMPv4 echo traffic with unusually uniform packet sizes and regular timing to external or explicitly unexpected destinations.

conn.log does not supply per-packet length distribution.

26. ICS Protocol Error / Exception Spike

`ics_protocol_error_spike`

Inputs: one of: modbus.log, dnp3.log, enip.log, cip.log, s7comm.log.

Detection: Detects elevated rates of ICS protocol error responses, including Modbus exceptions, DNP3 IIN error indications, EtherNet/IP/CIP general-status failures, and S7 error returns.

Log presence and field coverage must be verified at runtime. EtherNet/IP/CIP general-status detail is read from cip.log when ICSNPP-ENIP is loaded, falling back to enip.log alone when it is not.

Non-default Zeek requirement: enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek. | cip.log — ICSNPP package required: Requires the ICSNPP-ENIP package. ICSNPP-ENIP writes CIP service-code detail to a separate cip.log, not enip.log. Not present in stock Zeek. | s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek.

27. Modbus/DNP3 Write Operations Outside Allowed Paths

`ics_write_operations`

Inputs: one of: modbus.log, dnp3.log.

Detection: Detects Modbus write operations and DNP3 write/control operations outside configured source-to-destination paths, plus protocol function codes outside an optional approved profile.

Log presence and field coverage must be verified at runtime.

28. Internet-Exposed ICS Device / External Admin to PLC/HMI

`internet_exposed_ics`

Inputs: conn.log.

Detection: Identifies configured ICS assets communicating directly with public IP addresses and Internet-originated administrative access to PLC, RTU, HMI, BAS, historian, or other OT assets.

Log presence and field coverage must be verified at runtime.

29. IPv6 Traffic in OT Environment

`ipv6_traffic_ot`

Inputs: conn.log.

Detection: Identifies IPv6 communications observed in an OT/ICS environment that is explicitly documented as IPv4-only, including link-local, deprecated site-local, unique-local, multicast, and global IPv6.

Log presence and field coverage must be verified at runtime.

30. IRC Traffic Detected

`irc_traffic_detected`

Inputs: conn.log.

Detection: Identifies Internet Relay Chat traffic on standard IRC ports or on non-standard ports when Zeek explicitly identifies the service as IRC.

Log presence and field coverage must be verified at runtime.

31. JA3/JA3S Fingerprint Outliers

`ja3_fingerprint_outliers`

Inputs: ssl.log.

Detection: Detects rare TLS client/server JA3 or JA3S fingerprints that fall outside an established, stable fingerprint baseline for a configured network segment.

Stock TLS metadata does not establish JA3.

32. Kerberos AS-REP Roastable Accounts

`kerberos_asrep_roastable_accounts`

Inputs: kerberos.log; also used when present: conn.log.

Detection: Detects Kerberos accounts explicitly observed allowing AS authentication without pre-authentication, a condition associated with AS-REP roasting.

Default kerberos.log cannot prove absence of preauthentication. Disabled by default for this reason; enable only if the capture source is known to log pre-authentication fields.

DISABLED BY DEFAULT in eleVADR: stock Zeek cannot supply the evidence this module's claim needs, so it is switched off out of the box and produces no findings unless a non-stock capture pipeline is known to log the missing field.

33. Large Outbound POST/PUT to Unknown Hosts

`large_outbound_http_uploads`

Inputs: http.log; also used when present: conn.log.

Detection: Identifies substantial HTTP POST/PUT uploads from internal clients to external, non-approved destinations using explicit request-body length when available and conn.log orig_bytes only as a conservative fallback.

Log presence and field coverage must be verified at runtime.

34. LDAP Cleartext / Anonymous Bind / Cross-Segment

`ldap_cleartext_anonymous_cross_segment`

Inputs: any of (no single log mandatory): ldap.log, ldap_bind.log, ldap_search.log, conn.log.

Detection: Identifies LDAP anonymous or simple binds without TLS, cleartext LDAP on port 389, and LDAP communications that cross configured network segments, including OT-to-enterprise directory queries.

ldap_bind.log is not a stock log.

Non-default Zeek requirement: ldap_bind.log — Not a real Zeek log: Does not exist as a Zeek log. Bind mechanism/result is carried in ldap.log, not a separate ldap_bind.log.

35. LLMNR/NBT-NS/mDNS Poisoning Signals

`llmnr_nbtns_mdns_poisoning_signals`

Inputs: dns.log.

Detection: Detects unusually frequent multicast/broadcast name-resolution queries and suspicious LLMNR, NBT-NS, or mDNS responder patterns that can be consistent with name-resolution poisoning.

Log presence and field coverage must be verified at runtime. LLMNR and mDNS land in dns.log on stock Zeek; NBT-NS is not a DNS protocol and stock Zeek's DNS analyzer does not parse it, so NBT-NS detection here depends on a non-stock pipeline writing NBT-NS records into dns.log.

36. NetBIOS / SMBv1 Exposure

`netbios_smbv1_exposure`

Inputs: conn.log.

Detection: Identifies NetBIOS traffic on ports 137-139 crossing configured network-segment boundaries or involving public Internet addresses.

SMBv1 detection was removed from this module. It depended on smb_cmd.log, which is off by default in stock Zeek and known to drop commands on busy streams, and so was not a reliable basis for a finding. Only NetBIOS port-based detection remains.

37. New OT Conversation Pair (Comm-Matrix Drift)

`new_ot_conversation_pair`

Inputs: conn.log.

Detection: Builds a directed source-to-destination communication matrix during an OT baseline period and identifies new talker relationships that first appear after the baseline.

Log presence and field coverage must be verified at runtime.

38. New Service Emergence in OT (Baseline Drift)

`new_service_emergence_ot`

Inputs: conn.log.

Detection: Identifies services newly observed on configured OT hosts after an established baseline period, using Zeek service identification when available and protocol/port identity as a fallback.

Log presence and field coverage must be verified at runtime.

39. Niagara Fox Protocol Detected

`niagara_fox_detected`

Inputs: conn.log; also used when present: ssl.log.

Detection: Identifies Tridium Niagara Fox communications on TCP/1911 and Fox-over-SSL on TCP/4911, commonly used between JACE controllers and supervisory stations in building automation systems.

Log presence and field coverage must be verified at runtime.

40. NTP Sourced from Internet (Multi-Dest / OT)

`ntp_internet_multi_dest_ot`

Inputs: ntp.log.

Detection: Detects OT hosts using multiple distinct globally routable NTP servers instead of a centralized trusted time source, with optional trusted-server and external-server allow-list policy.

Log presence and field coverage must be verified at runtime.

41. NTP Source Drift

`ntp_source_drift`

Inputs: ntp.log.

Detection: Detects NTP clients that use servers outside explicitly trusted time infrastructure or change server after a capture baseline.

Log presence and field coverage must be verified at runtime.

42. OT Asset Gone Silent (Loss of Expected Periodic Comms)

`ot_asset_gone_silent`

Inputs: conn.log; optional (sharpens confidence/timing when present): modbus.log, dnp3.log, s7comm.log, enip.log, cip.log.

Detection: Learns regular OT host/conversation cadence during a baseline period and identifies expected periodic communications that cease for materially longer than their established interval. Uses per-message OT protocol-log timing when it is more granular than conn.log, so polling inside one persistent session is still visible.

Connection inactivity does not prove polling inactivity.

Non-default Zeek requirement: s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek. | enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek. | cip.log — ICSNPP package required: Requires the ICSNPP-ENIP package. ICSNPP-ENIP writes CIP service-code detail to a separate cip.log, not enip.log. Not present in stock Zeek.

43. OT Host Using External / Internet DNS Resolver

`ot_external_dns_resolver`

Inputs: dns.log.

Detection: Detects OT/control-system hosts sending DNS queries to public Internet resolvers or resolvers outside the configured OT boundary instead of approved internal DNS infrastructure.

Log presence and field coverage must be verified at runtime.

44. Certificate Long Validity / Self-Issued in OT Mgmt

`ot_management_certificate_risk`

Inputs: one of: ssl.log, x509.log.

Detection: Identifies long-validity and self-issued TLS certificates presented by configured OT management endpoints.

Log presence and field coverage must be verified at runtime.

45. OT Outbound Internet Connection (Any Protocol)

`ot_outbound_internet_any_protocol`

Inputs: conn.log.

Detection: Generic egress catch-all for sessions initiated by configured OT/control-system assets toward globally routable Internet destinations, irrespective of application protocol.

Log presence and field coverage must be verified at runtime.

46. OT Protocol Exposure

`ot_protocol_exposure`

Inputs: conn.log; optional (sharpens confidence/timing when present): modbus.log, dnp3.log, enip.log, bacnet.log, s7comm.log, mms.log, iec61850.log.

Detection: Identifies industrial control protocols observed outside configured control-system network segments, including Modbus, DNP3, EtherNet/IP, BACnet, IEC 61850 MMS, and S7comm.

Log presence and field coverage must be verified at runtime. IEC 61850 GOOSE/Sampled Values are L2 multicast, non-IP traffic and are invisible to an IP-oriented Zeek sensor; only MMS-over-TCP is covered here.

Non-default Zeek requirement: enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek. | bacnet.log — ICSNPP package required: Requires the ICSNPP-BACnet package (zkg install icsnpp-bacnet). Not present in stock Zeek. | s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek. | mms.log — ICSNPP package required: Requires the ICSNPP-MMS package (zkg install icsnpp-mms). Not present in stock Zeek. Covers MMS-over-TCP only; IEC 61850 GOOSE/Sampled Values are L2 multicast, non-IP traffic and are not obtainable from any Zeek log. | iec61850.log — Not obtainable from Zeek: Not obtainable from any Zeek log. GOOSE/Sampled Values are L2 multicast, non-IP traffic invisible to an IP-oriented Zeek sensor.

47. OT Protocol Role Reversal

`ot_protocol_role_reversal`

Inputs: conn.log.

Detection: Detects hosts that are established as OT protocol responders during the baseline and later begin originating that protocol toward peers.

Log presence and field coverage must be verified at runtime.

48. PLC Program / Logic Download or Firmware Update

`plc_program_logic_firmware_update`

Inputs: one of: modbus.log, s7comm.log, enip.log, dnp3.log.

Detection: Detects explicit ICS protocol operations associated with PLC program/logic transfer, firmware update, or DNP3 restart/file-transfer activity.

non-base/extension logs

Non-default Zeek requirement: s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek. | enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek.

49. PLC / RTU Peer Change

`plc_rtu_peer_change`

Inputs: conn.log.

Detection: Detects a new communication peer involving an authoritative PLC, RTU, controller, or IED after the capture baseline.

Log presence and field coverage must be verified at runtime.

50. Polling Cadence Disruption

`polling_cadence_disruption`

Inputs: conn.log; optional (sharpens confidence/timing when present): modbus.log, dnp3.log, s7comm.log, enip.log, cip.log.

Detection: Detects established periodic OT polling whose post-baseline interval shifts materially from its learned cadence. Uses per-message OT protocol-log timing when it is more granular than conn.log, so polling inside one persistent session is still visible.

Long-lived TCP sessions hide individual polling intervals.

Non-default Zeek requirement: s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek. | enip.log — ICSNPP package required: Requires the ICSNPP-ENIP package (zkg install icsnpp-enip). Not present in stock Zeek. | cip.log — ICSNPP package required: Requires the ICSNPP-ENIP package. ICSNPP-ENIP writes CIP service-code detail to a separate cip.log, not enip.log. Not present in stock Zeek.

51. Port / Host Scanning

`port_host_scanning`

Inputs: conn.log.

Detection: Identifies rapid connection attempts from a single source to many ports on one host (port scan) or to many hosts on the same service/port (host sweep).

Log presence and field coverage must be verified at runtime.

52. Protocol on Unexpected / High-Risk Port

`protocol_unexpected_high_risk_port`

Inputs: conn.log.

Detection: Identifies Zeek-recognized application protocols running on unexpected ports and traffic using ports with a strong historical association with backdoors, reverse shells, proxies, or command-and-control tooling.

Log presence and field coverage must be verified at runtime.

53. Public-to-Public Traffic in Capture

`public_to_public_traffic`

Inputs: conn.log.

Detection: Identifies traffic between two globally routable IP addresses when the monitored capture is explicitly expected to contain only internal ICS/OT communications.

Log presence and field coverage must be verified at runtime.

54. QUIC in OT Segments

`quic_ot_segments`

Inputs: conn.log; also used when present: quic.log.

Detection: Identifies QUIC/HTTP/3 traffic involving configured OT/control segments, with explicit protocol evidence preferred over lower-confidence UDP/443 heuristics.

quic.log is a preliminary/evolving analyzer only present on Zeek 6.1 and newer; older Zeek builds have no quic.log and this module falls back to a low-confidence UDP/443 heuristic from conn.log alone.

Non-default Zeek requirement: quic.log — Non-default configuration required: Stock Zeek only on version 6.1 or newer; not present on older Zeek builds.

55. Unexpected RDP with NLA Disabled

`rdp_nla_disabled`

Inputs: rdp.log; also used when present: conn.log.

Detection: Identifies RDP sessions where protocol telemetry explicitly shows Network Level Authentication (NLA/CredSSP) was not used, plus lower-severity cases where NLA is explicitly not required.

Log presence and field coverage must be verified at runtime.

56. Remote Access Session Anomaly

`remote_access_session_anomaly`

Inputs: conn.log.

Detection: Detects authorized remote-access sources that fan out to unusual OT targets after a baseline.

Log presence and field coverage must be verified at runtime.

57. VNC / Remote Access Tool Exposure

`remote_access_tool_exposure`

Inputs: conn.log; also used when present: vnc.log.

Detection: Identifies VNC, pcAnywhere, TeamViewer, AnyDesk, Radmin, and configured remote-access tools, with higher priority for traffic involving OT/ICS segments or public Internet peers.

Log presence and field coverage must be verified at runtime.

Non-default Zeek requirement: vnc.log — Add-on package required: Requires a separate Zeek package (not in stock Zeek).

58. Rogue DHCP Server / DHCP Activity on Static OT Segment

`rogue_dhcp_static_ot`

Inputs: dhcp.log.

Detection: Detects DHCP offers or lease activity from unapproved servers, competing offers from multiple DHCP servers, and DHCP request/offer/lease activity on explicitly static or DHCP-prohibited OT segments.

Log presence and field coverage must be verified at runtime.

59. S7comm Unauthorized Write/Stop PLC

`s7comm_unauthorized_write_stop`

Inputs: s7comm.log.

Detection: Detects explicit Siemens S7comm write/download/control operations and PLC STOP commands, with optional policy checks for approved source-to-destination control paths.

Requires S7 function-code evidence.

Non-default Zeek requirement: s7comm.log — ICSNPP package required: Requires the ICSNPP-S7comm package (zkg install icsnpp-s7comm). Not present in stock Zeek.

60. Service Disappearance / Replacement

`service_disappearance_replacement`

Inputs: conn.log.

Detection: Detects OT endpoints where a baseline service disappears and a different service appears after the baseline.

Log presence and field coverage must be verified at runtime.

61. SMB Admin-Share Access (IPC$/ADMIN$)

`smb_admin_share_access`

Inputs: any of (no single log mandatory): smb_mapping.log, smb_files.log, smb_cmd.log, dce_rpc.log, conn.log.

Detection: Identifies explicit SMB access to IPC$ or ADMIN$ and highlights repeated or multi-target patterns that can be consistent with remote administration or lateral movement.

Avoid relying on nonexistent smb.log. dce_rpc.log svcctl/atsvc operations are used as corroborating evidence only, not independent proof of compromise.

Non-default Zeek requirement: smb_cmd.log — Stock Zeek, disabled by default: Ships with stock Zeek but is disabled by default; enabling it (SMB::logged_smb_cmds) exposes a debug-oriented log known to drop commands on busy streams and omit negotiate/session-setup.

62. SMB Signing Disabled or NTLMv1 Detected

`smb_signing_disabled_ntlmv1`

Inputs: any of (no single log mandatory): smb_mapping.log, smb_files.log, smb_cmd.log, ntlm.log, conn.log.

Detection: Identifies SMB sessions explicitly observed without message signing or with signing disabled/not required, and NTLM authentication records that explicitly identify NTLMv1.

Default SMB/NTLM logs cannot establish full claim. Disabled by default for this reason; enable only if the capture source is known to log signing/NTLM-version fields.

Non-default Zeek requirement: smb_cmd.log — Stock Zeek, disabled by default: Ships with stock Zeek but is disabled by default; enabling it (SMB::logged_smb_cmds) exposes a debug-oriented log known to drop commands on busy streams and omit negotiate/session-setup.

DISABLED BY DEFAULT in eleVADR: stock Zeek cannot supply the evidence this module's claim needs, so it is switched off out of the box and produces no findings unless a non-stock capture pipeline is known to log the missing field.

63. SNMP Write / Set to OT Devices

`snmp_write_ot_devices`

Inputs: snmp.log.

Detection: Detects SNMP SET operations or use of explicitly configured writable SNMPv1/v2c communities against OT devices, controllers, or managed network infrastructure beyond ordinary read-only polling.

Log presence and field coverage must be verified at runtime.

64. SOCKS / Open Proxy Behavior

`socks_open_proxy_behavior`

Inputs: one of: socks.log, http.log; also used when present: conn.log.

Detection: Identifies hosts acting as SOCKS or HTTP CONNECT proxy endpoints using explicit protocol telemetry, without inferring proxy behavior from common ports alone.

Log presence and field coverage must be verified at runtime.

65. TCP Reset / Connection-Abort Surge

`tcp_reset_abort_surge`

Inputs: conn.log.

Detection: Detects bursts of reset/rejected/aborted TCP sessions against the same destination.

Log presence and field coverage must be verified at runtime.

66. TLS Certificate Anomalies

`tls_certificate_anomalies`

Inputs: any of (no single log mandatory): ssl.log, x509.log.

Detection: Identifies self-signed, expired, hostname-mismatched, weak-key, and unknown/untrusted-CA TLS certificates using Zeek ssl.log and x509.log when available.

Log presence and field coverage must be verified at runtime.

67. TLS SNI Mismatch / Certificate Reuse Across Hosts

`tls_sni_certificate_reuse`

Inputs: one of: ssl.log, x509.log.

Detection: Detects TLS sessions whose SNI/server identity does not match the presented certificate and certificates reused broadly across unrelated server hosts/domains.

Log presence and field coverage must be verified at runtime.

68. Unexpected DHCP Server

`unexpected_dhcp_server`

Inputs: dhcp.log.

Detection: Detects DHCP server responses from sources outside the explicitly configured trusted DHCP infrastructure.

Log presence and field coverage must be verified at runtime.

69. Unexpected Multicast Behavior

`unexpected_multicast_behavior`

Inputs: conn.log.

Detection: Detects OT hosts initiating multicast traffic to groups outside explicitly allowed multicast destinations.

Log presence and field coverage must be verified at runtime.

70. Unknown / Rogue Devices

`unknown_rogue_devices`

Inputs: any of (no single log mandatory): conn.log, dhcp.log, arp.log.

Detection: Identifies locally observed IP or MAC addresses that are not present in the supplied asset inventory.

Log presence and field coverage must be verified at runtime.

Non-default Zeek requirement: arp.log — Custom script required: No stock arp.log. Requires a custom ARP-logging script, and the sensor must sit on the Layer-2 broadcast segment to see the traffic at all.

71. Unusual Outbound Data Volume

`unusual_outbound_data_volume`

Inputs: conn.log.

Detection: Identifies connections with unusually large Zeek orig_bytes leaving local networks, especially to external destinations or across explicitly configured trust boundaries.

Log presence and field coverage must be verified at runtime.

72. UPnP / SSDP Exposure & IGD Port Mapping

`upnp_ssdp_igd_port_mapping`

Inputs: one of: upnp.log, upnp_igd.log, ssdp.log; also used when present: conn.log.

Detection: Detects explicit UPnP Internet Gateway Device port-mapping activity and unexpected SSDP exposure while avoiding port-only inference from UDP/1900.

non-base/extension logs

Non-default Zeek requirement: upnp.log — Add-on package required: Requires a separate Zeek package (not in stock Zeek). | upnp_igd.log — Add-on package required: Requires a separate Zeek package (not in stock Zeek). | ssdp.log — Add-on package required: Requires a separate Zeek package (not in stock Zeek).

73. VLAN Tag Mismatch / Double-Tag Indicators

`vlan_tag_mismatch_double_tag`

Inputs: conn.log.

Detection: Detects explicit 802.1Q VLAN IDs in Zeek conn.log that fall outside configured VLAN policy, and double-tagged/Q-in-Q connections that may indicate VLAN hopping or unexpected trunk/provider tagging.

conn.log carries no vlan/inner_vlan fields by default; the vlan-logging.zeek policy script must be explicitly loaded (redef) to add them. SPAN/tap may also strip 802.1Q tags before capture, which makes double-tag detection impossible regardless of the logging policy.

Non-default Zeek requirement: conn.log — Non-default configuration required: Requires a non-default Zeek policy script to be loaded (e.g. redef'ing vlan-logging.zeek) before the needed fields appear in the log.

74. Weak or Broken TLS/SSL

`weak_broken_tls_ssl`

Inputs: any of (no single log mandatory): ssl.log, x509.log.

Detection: Identifies deprecated SSL/TLS versions, weak cipher suites, and weak certificate signature algorithms when the relevant Zeek logs are present.

Log presence and field coverage must be verified at runtime.

75. Weird.log Protocol Violations

`weird_protocol_violations`

Inputs: weird.log; also used when present: conn.log.

Detection: Analyzes Zeek weird.log for malformed packets, checksum anomalies, unexpected protocol states, and other protocol-specification violations.

Log presence and field coverage must be verified at runtime.

Source baseline: eleVADR Dataset 18 real-site validation checkpoint; 75 registered modules. This reference summarizes the detector source code and module metadata as of 2026-11-03, cross-checked against the Zeek Log-Source Validation review and a full source-code audit of every module.